Polynomial Lerch Selectors

Endpoint obstructions, resonant spectral collapse,
and Stieltjes confluence

Research continuation prepared for
Vladimir Reshetnikov's ProveIt project

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Abstract

We answer the polynomial-selector question posed in the incoming *Affine Gamma Polarization* report for every positive integer interval length and every nonzero twist in the closed unit disk. A finite vector of paired endpoint derivatives is exactly the support of the shifted Lerch functions in the transform, modulo a specified finite exponential remainder. Off resonance, the unique single-order weights are classical Frobenius–Euler polynomials; at resonance they are Bernoulli polynomials, with an arbitrary constant. Functional independence over $\mathbb{C}(s)$ makes this a classification of analytic functions, not merely of formal coordinates.

Eulerian polynomials give a sharp endpoint obstruction: off resonance a nonzero single-order weight cannot vanish to order two at both endpoints. At resonance, double zeros occur precisely for the even Bernoulli weights of degree at least four, and triple zeros never occur. An explicit finite mixture replaces the impossible scalar confluence; its number of spectral coordinates drops exactly from $r+1$ to one. We derive every removable-integer spectral jet through finite generalized-harmonic coefficients and obtain Euler-weighted Gamma integrals with one odd zeta value. A joint holomorphic renormalization at the Hurwitz pole supplies every mixed twist/Stieltjes jet and anchored shift primitives.

Classical polynomial bases and the classical Lerch expansion are credited explicitly. The contribution is their exact selector classification, endpoint and confluence consequences, and the resulting identity calculus in the stated source programme. We also solve an explicit coefficient problem in arXiv:2609.13796v1 and correct three normalization/sign statements relative to its defining series. The package includes 969 finite exact assertions and 33 high-precision diagnostics. These are not interval certificates or proof-assistant verification. No global priority claim or resolution of the S_6, revised S_8, or general unequal-tail prefix problems is made.

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1 The source question and the scope of the answer

The canonical ProveIt manuscript separates its proven identities, experimental candidates, and pending analytic integrations [1]. The incoming *Affine Gamma Polarization* report extends the Gamma square sector to weighted Hurwitz and Lerch transforms. Its Section 11, questions 4 and 5, asks for a functional classification of polynomial weights on integer intervals that leave one shifted zeta or polylogarithm order plus an elementary remainder, and asks how the twisted answer approaches the Bernoulli selectors [2]. These are the specific continuation questions addressed here. We do not repeat that report's Gamma normal-form theorem.

The phrase "one order" needs a coefficient field. At a single numerical value of s, identities between constants could obscure the issue. With arbitrary meromorphic coefficients every nonzero function could trivially be called a selector. We instead allow coefficients in $\mathbb{C}(s)$ and fix the finite exponential remainder arising from integer translation. This is the same functional, rather than numerical, standard used in the predecessor's Bernoulli classification.

Let $a > 0$, $L \geq 1$ be an integer, $P \in \mathbb{C}[u]$, and $0 < |z| \leq 1$. The central transform is

$$\mathcal{T}_P(z, s; a, L) = \int_0^L P(u)\Phi(z, s, a+u)\,\mathrm{d}u, \qquad \Phi(z, s, x) = \sum_{n=0}^{\infty} \frac{z^n}{(n+x)^s}. \tag{1.1}$$

For $|z| < 1$ the series is entire in s. At $|z| = 1$ the identities are first proved for $\Re s > 1$ and then continued. For $z \neq 1$ the continued transform is entire; for $z = 1$ it is a weighted Hurwitz-zeta integral with only a possible pole at $s = 1$. Positive bases use their real logarithms. No branch ambiguity in $(n+x)^{-s}$ is introduced.

Theorem 3.1 solves the exact selector problem. Theorem 4.2 proves the endpoint obstruction and its resonant exception. Theorem 5.2 describes constructive confluence with an exact support count. Theorems 6.1 and 9.1 give the spectral and Stieltjes calculi. The concrete Gamma and quartic-weight identities in Sections 7 and 8 are useful normalization tests and applications.

Attribution and novelty scope. Frobenius–Euler polynomials, their Appell property and their endpoint basis formula are classical; see Kim and Kim [3]. Eulerian real-rootedness is classical as well; modern interlacing proofs include Chow [4]. Hurwitz differentiation, Lerch translation, and the logarithmic Lerch expansion are classical [6, 5]. The predecessor already proves a single-zeta Bernoulli basis on $[1, 3]$ and weighted Stieltjes formulas. We give self-contained proofs where short, but do not claim these ingredients as discoveries. The results below resolve the stated source questions in an explicit domain and exhibit consequences not established in the inspected source sections. A finite literature search cannot establish global priority.

2 Paired endpoint derivatives are the exact spectrum

Write

$$F_j(s) = \prod_{k=1}^{j+1}(s-k), \qquad j \geq 0, \qquad \lambda = z^L. \tag{2.1}$$

For a polynomial of degree D, set

$$c_j(P; \lambda, L) = P^{(j)}(L) - \lambda P^{(j)}(0), \qquad 0 \leq j \leq D. \tag{2.2}$$

The elementary remainder space in the next results is

$$\mathcal{E}_{a,L} = \operatorname{span}_{\mathbb{C}(s)}\{(a+\ell)^{-s} : 0 \le \ell < L\}. \tag{2.3}$$

The twist is fixed while taking this span. Coefficients may depend on that fixed twist and on the fixed interval parameters.

Proposition 2.1 (Finite endpoint transform)**.** *As meromorphic functions of s,*

$$\mathcal{T}_P(z,s;a,L) = -\sum_{j=0}^{D} \frac{P^{(j)}(L)\Phi(z,s-j-1,a+L) - P^{(j)}(0)\Phi(z,s-j-1,a)}{F_j(s)} \tag{2.4}$$

$$= -\lambda^{-1}\sum_{j=0}^{D} \frac{c_j(P;\lambda,L)}{F_j(s)}\Phi(z,s-j-1,a) + \mathcal{R}_P(z,s;a,L), \tag{2.5}$$

where the explicitly elementary remainder is

$$\mathcal{R}_P = \lambda^{-1}\sum_{j=0}^{D} \frac{P^{(j)}(L)}{F_j(s)} \sum_{\ell=0}^{L-1} z^{\ell}(a+\ell)^{j+1-s}. \tag{2.6}$$

All apparent poles other than the possible Hurwitz pole at $s=1$ cancel in the complete expression.

Proof. Termwise differentiation in an initial convergence domain gives $\partial_x\Phi(z,s,x) = -s\Phi(z,s+1,x)$. Successive integration by parts terminates after $D+1$ steps. At each step the antiderivative contributes the next factor $-(s-k)^{-1}$ while integration by parts contributes another minus sign to the continuing integral. Thus the boundary sum has the uniform minus sign in (2.4). Equivalently, $(-1)^j/(1-s)_{j+1} = -1/F_j(s)$.

Integer translation of the absolutely convergent series gives

$$\Phi(z,\sigma,a) = z^L\Phi(z,\sigma,a+L) + \sum_{\ell=0}^{L-1} z^{\ell}(a+\ell)^{-\sigma}.$$

Substitution proves (2.5). Analytic continuation then proves the formulas everywhere they are asserted. For $z \neq 1$ on the unit circle, the Mellin representation with denominator $1-ze^{-t}$ continues the function to an entire function of s: near zero its numerator is analytic, and the zeros of $1/\Gamma(s)$ cancel the Mellin poles at nonpositive integers. The corresponding assertion at $z=1$ is the usual Hurwitz continuation. Integration over a compact positive interval preserves these properties. □

Lemma 2.2 (Functional independence)**.** *For distinct nonnegative integers $j_1,\dots,j_m$, the functions*

$$\Phi(z,s-j_1-1,a),\dots,\Phi(z,s-j_m-1,a), \quad (a+\ell)^{-s}\ (0 \le \ell < L)$$

are linearly independent over $\mathbb{C}(s)$.

Proof. Clear rational denominators in a putative relation. For real s sufficiently large, expand it into a Dirichlet series

$$\sum_{n\ge 0} Q_n(s)(a+n)^{-s} = 0.$$

The polynomials $Q_n(s)$ have uniformly bounded degree in s. For $n \geq L$ they have the form

$$Q_n(s) = z^n \sum_{\nu=1}^{m} p_\nu(s)(a+n)^{j_\nu+1},$$

where the p_ν are fixed polynomials. The finitely many smaller indices contain the elementary coefficients as well. There are constants C, M, K for which $|Q_n(s)| \leq C(1+s)^M(1+n)^K$ on the positive real axis.

If one coefficient polynomial is nonzero, choose the least such index n_0. Divide the identity by $(a+n_0)^{-s}$. The tail tends to zero exponentially, up to a fixed polynomial in s. Indeed, for large s one can split the exponent in half and bound it by

$$C'(1+s)^M \left(\frac{a+n_0}{a+n_0+1}\right)^{s/2} \sum_{n>n_0} (1+n)^K \left(\frac{a+n_0}{a+n}\right)^{s/2},$$

where the final sum is uniformly bounded once $s/2 > K+2$. A nonzero polynomial $Q_{n_0}(s)$ cannot decay exponentially. Thus every Q_n is zero.

For infinitely many $n \geq L$, division by $z^n \neq 0$ gives $\sum_\nu p_\nu(s)(a+n)^{j_\nu+1} = 0$. A polynomial in $a+n$ with infinitely many roots is identically zero; hence each $p_\nu = 0$. The remaining finite exponential coefficients are then zero too. □

Corollary 2.3 (No hidden spectral cancellation)**.** *Modulo $\mathcal{E}_{a,L}$, the transform has precisely the shifted orders $s-j-1$ for which $c_j(P;\lambda,L) \neq 0$. This support is unique.*

For positive integer $a = A$, the relation

$$\Phi(z,\sigma,A) = z^{-A}\left(\mathrm{Li}_\sigma(z) - \sum_{n=1}^{A-1} \frac{z^n}{n^\sigma}\right) \tag{2.7}$$

turns this into an exact polylogarithmic statement. The elementary space is then enlarged, explicitly, to $\mathrm{span}_{\mathbb{C}(s)}\{n^{-s} : 1 \leq n < A+L\}$. The same least-index argument proves independence in that form.

3 Classification off and at resonance

We call $\lambda = z^L = 1$ *resonance*. This concerns the interval translation, not only the point $z = 1$. Nontrivial roots of unity can therefore be resonant, while their Lerch functions remain entire in s.

Define the classical Frobenius–Euler polynomials by

$$\frac{(1-\lambda)e^{xt}}{e^t - \lambda} = \sum_{d=0}^{\infty} \mathsf{H}_d(x;\lambda)\frac{t^d}{d!}, \qquad \lambda \neq 1. \tag{3.1}$$

They are monic and satisfy

$$\partial_x \mathsf{H}_d(x;\lambda) = d\mathsf{H}_{d-1}(x;\lambda), \tag{3.2}$$

$$\mathsf{H}_d(1;\lambda) - \lambda \mathsf{H}_d(0;\lambda) = (1-\lambda)\mathbf{1}_{d=0}. \tag{3.3}$$

These identities follow immediately from the generating function. The notation H_d is distinguished from generalized harmonic numbers $H_n^{(r)}$ used later.

Theorem 3.1 (All single-order polynomial selectors)**.** *Fix $a > 0$, integer $L \geq 1$, and $0 < |z| \leq 1$. Let $N \geq 1$. A polynomial P has a nonzero single-order representation*

$$\mathcal{T}_P(z, s; a, L) = A(s)\Phi(z, s - N, a) + E(s), \qquad 0 \neq A \in \mathbb{C}(s), \quad E \in \mathcal{E}_{a,L}, \tag{3.4}$$

if and only if one of the following holds.

If $\lambda \neq 1$, then

$$P(u) = CL^{N-1}\mathsf{H}_{N-1}(u/L; \lambda), \qquad C \neq 0. \tag{3.5}$$

In particular $\deg P = N - 1$, and $A(s) = -\lambda^{-1}(1 - \lambda)C(N - 1)!/F_{N-1}(s)$. No nonzero polynomial has an elementary-only transform.

If $\lambda = 1$, then

$$P(u) = CL^N B_N(u/L) + C_0, \qquad C \neq 0, \tag{3.6}$$

where B_N is the Bernoulli polynomial. Now $\deg P = N$ and $A(s) = -CLN!/F_{N-1}(s)$. The elementary-only polynomials are exactly the constants.

Proof. For $\lambda \neq 1$, (3.2)–(3.3) show that the polynomial $L^d\mathsf{H}_d(u/L; \lambda)$ has only one nonzero endpoint coordinate, namely $c_d = (1 - \lambda)d!$. Since these monic polynomials form a basis, every P of degree at most D has the classical endpoint expansion

$$P(u) = \sum_{d=0}^{D} \frac{c_d(P; \lambda, L)L^d}{(1 - \lambda)d!}\mathsf{H}_d(u/L; \lambda). \tag{3.7}$$

Corollary 2.3 proves that a single order is possible exactly when one coefficient in this expansion survives. The highest endpoint coordinate of a nonzero degree-D polynomial is $(1 - \lambda)P^{(D)}$, so its sole coordinate must have $d = D$. This proves the nonresonant assertion.

At resonance use $B_n(1) - B_n(0) = \mathbf{1}_{n=1}$, which follows from $te^{xt}/(e^t - 1)$. It gives the anchored expansion

$$P(u) = P(0) + \sum_{N=1}^{D} \frac{c_{N-1}(P; 1, L)L^{N-1}}{N!}(B_N(u/L) - B_N(0)). \tag{3.8}$$

Only the coordinate $j = N - 1$ is nonzero for a nonconstant basis term. The kernel consists of constants: alternatively, a polynomial with all paired derivatives zero must be constant, by looking successively at its highest nonconstant degree. The independence lemma again turns this basis statement into the functional classification. □

Remark 3.2 (A finite interpolation problem, with a singular limit). On the monomial basis $1, u, \ldots, u^D$, the endpoint map $P \mapsto (c_0, \ldots, c_D)$ is upper triangular and has determinant

$$(1 - \lambda)^{D+1} \prod_{j=0}^{D} j!. \tag{3.9}$$

At $\lambda = 1$ it has rank D, with constants as kernel. The minor with rows $c_0, \ldots, c_{D-1}$ and columns $u, \ldots, u^D$ has

$$\det = L^D \prod_{j=1}^{D} j! \neq 0. \tag{3.10}$$

Thus the degree shift in Theorem 3.1 is an exact change of rank, not a numerical instability.

4 Endpoint zeros and the Eulerian obstruction

The source's Gamma-derived weights vanish with their first derivatives at both endpoints. Such a condition is not innocuous under twisting. We now classify precisely when it is compatible with one order.

Define Eulerian polynomials A_d for $d \geq 1$ by

$$\sum_{n=1}^{\infty} n^d \lambda^n = \frac{\lambda A_d(\lambda)}{(1-\lambda)^{d+1}}, \qquad |\lambda| < 1. \tag{4.1}$$

Here $A_1 = 1$, $A_2 = 1 + \lambda$, and $A_3 = 1 + 4\lambda + \lambda^2$. Differentiating the generating series with $\lambda\partial_\lambda$ gives

$$A_d = (1 + (d-1)\lambda)A_{d-1} + \lambda(1-\lambda)A'_{d-1}. \tag{4.2}$$

Lemma 4.1 (The endpoint Eulerian polynomial)**.** *For $d \geq 1$ and $\lambda \neq 1$,*

$$\mathsf{H}_d(0;\lambda) = \frac{(-1)^d A_d(\lambda)}{(1-\lambda)^d}. \tag{4.3}$$

Every zero of A_d is simple and negative. Consecutive A_d, A_{d-1}, for $d \geq 2$, have no common zero.

Proof. For $|\lambda| < 1$, write the left side of (3.1) at $x = 0$ as $(1-\lambda)\sum_{n\geq 0} \lambda^n e^{-(n+1)t}$. Taking its dth derivative at zero and using (4.1) proves (4.3). Rational continuation proves it for every $\lambda \neq 1$.

For completeness, real-rootedness follows directly by induction from (4.2). The polynomial A_{d-1} is monic of degree $d-2$, has value one at zero, and, inductively, has simple negative roots $r_1 < \cdots < r_{d-2}$. At each such root,

$$A_d(r_i) = r_i(1-r_i)A'_{d-1}(r_i) \neq 0.$$

These values alternate in sign. There is one zero of A_d in each successive gap, one between r_{d-2} and zero, and one to the left of r_1, by comparing its monic leading sign at $-\infty$. These $d-1$ distinct negative zeros exhaust its degree. The base $A_2 = 1 + \lambda$ starts the argument; A_1 has no roots. The same displayed identity proves the asserted absence of common zeros. □

Theorem 4.2 (Sharp endpoint multiplicity obstruction)**.** *A nonzero polynomial single-order selector has the following endpoint possibilities.*

For $\lambda \neq 1$, its degree-d weight can vanish at both endpoints only if $d \geq 2$ and $A_d(\lambda) = 0$. These zeros of the weight are then simple at both endpoints. In particular no nonzero single-order selector satisfies

$$P(0) = P'(0) = P(L) = P'(L) = 0. \tag{4.4}$$

For real $0 < z < 1$, even simultaneous endpoint values zero are impossible.

At $\lambda = 1$, a single-order selector can satisfy (4.4) if and only if it is a nonzero multiple of

$$B_{2r}(u/L) - B_{2r}, \qquad r \geq 2. \tag{4.5}$$

Both endpoint zeros then have order exactly two. No nonzero single-order selector in either regime has zeros of order three at both endpoints.

Proof. For $d \geq 1$, (3.3) makes the two endpoint values proportional by the nonzero factor λ. Their vanishing is therefore equivalent to $A_d(\lambda) = 0$. The cases $d = 0, 1$ have no vanishing endpoint value. If $d \geq 2$ and the value is zero, the Appell identity makes the derivative proportional to $\mathsf{H}_{d-1}(0; \lambda)$, which is nonzero by the lemma. The derivative at the other endpoint is again multiplied by λ, so it is also nonzero. If $0 < z < 1$, then $\lambda > 0$, where A_d has no root.

At resonance the classification permits only $CL^N B_N(u/L) + C_0$. For $N = 1$ its two endpoint values differ, so they cannot both be zero. For $N \geq 2$, requiring them both to vanish fixes the constant and gives a multiple of $B_N(u/L) - B_N$. Its derivative vanishes at both endpoints exactly when $B_{N-1}(0) = B_{N-1}(1) = 0$, namely when $N \geq 4$ is even. Its second derivative is a nonzero multiple of B_{N-2}, a nonzero even Bernoulli number. The zeros have exactly order two. □

Example 4.3 (A nontrivial algebraic twist)**.** For $L = 1$, $z = \sqrt{3} - 2$, and $d = 3$, the identity $A_3(z) = 0$ gives the exact weight

$$\mathsf{H}_3(u; \sqrt{3} - 2) = u(u - 1)\left(u - \frac{1 + \sqrt{3}}{2}\right). \tag{4.6}$$

It selects only $\mathrm{Li}_{s-4}(\sqrt{3} - 2)$ modulo the elementary remainder. Both endpoint zeros are simple. This distinguishes "endpoint vanishing" from the stronger and impossible off-resonant condition (4.4).

5 Constructive confluence and exact spectral collapse

The Frobenius–Euler family does not approach the Bernoulli family by an innocuous scalar normalization. First we make that obstruction precise, and then give an exact finite replacement.

Proposition 5.1 (No nonconstant scalar limit)**.** *Fix $d \geq 1$. Suppose that $C(\lambda)\mathsf{H}_d(u; \lambda)$ has a finite coefficientwise limit as $\lambda \to 1$. That limit is constant. The statement also holds after the fixed scaling $u \mapsto u/L$.*

Proof. The recurrence (4.2) gives $A_d(1) = d!$. Consequently

$$\mathsf{H}_d(0; \lambda) \sim \frac{(-1)^d d!}{(1 - \lambda)^d}.$$

Boundedness of the constant coefficient forces $C(\lambda) = O(|1 - \lambda|^d)$. By the Appell property, the coefficient of u^j is $\binom{d}{j}\mathsf{H}_{d-j}(0; \lambda)$, with pole order at most $d - j$. After multiplication by $C(\lambda)$ every coefficient with $j \geq 1$ tends to zero. The constant coefficient may have a finite limit. □

This proposition concerns one fixed degree and scalar multiplication only. It does not forbid finite combinations of different degrees, subtraction of lower-degree counterterms, or unbounded-degree limits. The finite combination below is especially sparse.

Theorem 5.2 (Minimal mixing for the fixed even Bernoulli weight)**.** *Let $N = 2r \geq 4$ and $W_N(u) = B_N(u) - B_N$. For $z \neq 1$,*

$$\boxed{W_N(u) = \sum_{\substack{2 \leq j \leq N \\ j \text{ even}}} \binom{N}{j} B_{N-j}\mathsf{H}_j(u; z) + \frac{N(1 + z)}{2(1 - z)}\mathsf{H}_{N-1}(u; z).} \tag{5.1}$$

For the transform on $[0,1]$*, the number of distinct shifted Lerch orders modulo the elementary remainder is exactly*

$$\begin{cases} r+1, & z \notin \{0,1,-1\}, \quad |z| \le 1, \\ r, & z = -1, \\ 1, & z = 1. \end{cases} \tag{5.2}$$

The mixture in (5.1) *is unique in the Frobenius–Euler basis, and no representation of this fixed transform can use fewer distinct shifted orders in the declared coefficient system. On* $[0,L]$ *with* $W_N(u/L)$ *the same count holds with* z *replaced by* $\lambda = z^L$*.*

Proof. Put $b_j = W_N^{(j)}(1)$. The even nonzero endpoint derivatives are

$$b_j = \frac{N!}{(N-j)!} B_{N-j}, \qquad j = 2, 4, \dots, N. \tag{5.3}$$

There is also the odd derivative $b_{N-1} = N!/2$; every other odd derivative is zero, as are b_0, b_1. For even j, the left and right endpoint derivatives coincide. For $j = N-1$ they are opposite. Thus

$$c_j = (1-z)b_j \quad (j \text{ even}), \qquad c_{N-1} = (1+z)N!/2, \tag{5.4}$$

and all other coordinates vanish. Substitute these coordinates into (3.7) to obtain (5.1). Every even Bernoulli number in (5.3), including B_0, is nonzero. The independence lemma therefore gives the exact count (5.2), including its specializations at $z = \pm 1$. Scaling the interval multiplies b_j by L^{-j} and changes the paired endpoint factor to z^L, without changing the count. □

For example,

$$u^2(1-u)^2 = \mathsf{H}_4(u;z) + \mathsf{H}_2(u;z) + \frac{2(1+z)}{1-z}\mathsf{H}_3(u;z). \tag{5.5}$$

At $z = -1$ this is simply $E_4(u) + E_2(u)$, where E_d is the ordinary Euler polynomial. As $z \to 1$, the polynomial on the left stays fixed while the individual terms on the right need not have finite limits. This is the precise cancellation missed by a scalar confluence argument.

For $a = 1$, an explicit version of the transform is

$$\begin{aligned} \int_0^1 W_N(u)\Phi(z,s,1+u)\,\mathrm{d}u = & -\frac{N!(1+z)}{2z^2 F_{N-1}(s)}\,\mathrm{Li}_{s-N}(z) \\ & -\frac{1-z}{z^2}\sum_{\substack{2\le j\le N \\ j \text{ even}}} \frac{b_j\,\mathrm{Li}_{s-j-1}(z)}{F_j(s)} + \frac{1}{z}\sum_{j=2}^{N}\frac{b_j}{F_j(s)}. \end{aligned} \tag{5.6}$$

It follows directly from (2.5). Its limit at $z = 1$ is initially justified for $\Re s > 1$. At $s = 1$ the limiting Hurwitz pole requires the compensation in Section 9; it must not be suppressed by substituting into individual summands.

6 Removable integer orders and harmonic jet coefficients

The endpoint denominators vanish at positive integers, even when the integral is entire in s. The selector classification allows these limits to be evaluated using derivatives at just one polylogarithm order. We give a complete finite formula rather than leaving a removable $0/0$ expression.

For $z \neq 0, 1$ in the closed unit disk define

$$T_d(z,s) = \int_0^1 \mathsf{H}_d(u;z)\Phi(z,s,1+u)\,\mathrm{d}u, \tag{6.1}$$

$$\Pi_d(s) = \prod_{k=1}^{d+1}(s-k), \tag{6.2}$$

$$R_d(s,z) = \sum_{j=0}^{d} \mathsf{H}_d^{(j)}(1;z) \prod_{k=j+2}^{d+1}(s-k). \tag{6.3}$$

The superscript (j) in (6.3) differentiates the polynomial variable. Empty products equal one. The endpoint formula and $\Phi(z,\sigma,1) = z^{-1}\operatorname{Li}_\sigma(z)$ yield

$$\Pi_d(s)T_d(z,s) = \frac{R_d(s,z)}{z} - \frac{(1-z)d!}{z^2}\operatorname{Li}_{s-d-1}(z) =: N_d(s,z). \tag{6.4}$$

The numerator is entire in s and vanishes at each $m = 1, \ldots, d+1$.

Theorem 6.1 (Every integer spectral jet of a selector)**.** *For $1 \leq m \leq d+1$, put*

$$p_{d,m} = (-1)^{d+1-m}(m-1)!(d+1-m)!, \tag{6.5}$$

and define the analytic germ

$$E_{d,m}(\varepsilon) = \prod_{\substack{1\leq k\leq d+1\\ k\neq m}} \left(1 + \frac{\varepsilon}{m-k}\right)^{-1} \tag{6.6}$$

$$= \exp\left(\sum_{r\geq 1} \frac{(-1)^r H_{m-1}^{(r)} + H_{d+1-m}^{(r)}}{r}\varepsilon^r\right). \tag{6.7}$$

Here $H_n^{(r)} = \sum_{k=1}^n k^{-r}$ and $H_0^{(r)} = 0$. For every integer $h \geq 0$,

$$\boxed{\partial_s^h T_d(z,m) = \frac{h!}{p_{d,m}}[\varepsilon^{h+1}]E_{d,m}(\varepsilon)N_d(m+\varepsilon,z).} \tag{6.8}$$

In particular,

$$T_d(z,m) = \frac{z^{-1}\partial_s R_d(m,z) - (1-z)d!z^{-2}\,\partial_\nu \operatorname{Li}_\nu(z)|_{\nu=m-d-1}}{p_{d,m}}. \tag{6.9}$$

Thus an hth spectral derivative requires at most the first $h+1$ order derivatives at the single center $\nu = m-d-1$.

Proof. Factor the denominator at its simple zero:

$$\Pi_d(m+\varepsilon) = p_{d,m}\varepsilon E_{d,m}(\varepsilon)^{-1}.$$

Since $N_d(m,z) = 0$, division and Taylor coefficient extraction prove (6.8). Expanding $-\log(1+\varepsilon/(m-k))$ gives (6.7): positive differences contribute $(-1)^r H_{m-1}^{(r)}$, while negative differences contribute $H_{d+1-m}^{(r)}$. The constant term of $E_{d,m}$ is one, so (6.9) is the case $h = 0$. □

The coefficient rule is executable without symbolic limits. If $E_{d,m} = \sum_{n\ge 0} e_n \varepsilon^n$ and $A_r = (-1)^r H^{(r)}_{m-1} + H^{(r)}_{d+1-m}$, then

$$e_0 = 1, \qquad e_n = \frac{1}{n}\sum_{r=1}^{n} A_r e_{n-r}. \tag{6.10}$$

For example, $e_1 = A_1$ and $e_2 = (A_1^2 + A_2)/2$. The first derivative reads

$$\partial_s T_d(z,m) = \frac{\frac{1}{2}\partial_s^2 N_d(m,z) + A_1 \partial_s N_d(m,z)}{p_{d,m}}.$$

These are finite generalized-harmonic identities, not asymptotic expansions in a large index.

Example 6.2 (Degrees zero and one)**.** The first two formulas are

$$T_0(z,s) = \frac{z^{-1} - (1-z)z^{-2}\operatorname{Li}_{s-1}(z)}{s-1}, \tag{6.11}$$

$$T_1(z,s) = \frac{(1+z-zs)/(z(1-z)) - (1-z)z^{-2}\operatorname{Li}_{s-2}(z)}{(s-1)(s-2)}. \tag{6.12}$$

Here $\mathsf{H}_0 = 1$ and $\mathsf{H}_1(u;z) = u - (1-z)^{-1}$. At the first resonance,

$$\int_1^2 \Phi(z,1,x)\,\mathrm{d}x = -\frac{1-z}{z^2}\,\partial_\nu \operatorname{Li}_\nu(z)\big|_{\nu=0}. \tag{6.13}$$

The negative powers of z in these reductions have removable limits at $z = 0$ when the original polynomial family itself is regular there. The classification deliberately excludes $z = 0$, where the infinite Dirichlet tail disappears and the independence lemma no longer applies.

7 Euler weights, Gamma primitives, and one odd zeta

At $z = -1$, (3.1) becomes $2e^{ut}/(e^t+1)$, so $\mathsf{H}_d(u;-1) = E_d(u)$ is the ordinary Euler polynomial. Write

$$G(u) = \log\Gamma(1+u/2) - \log\Gamma((1+u)/2). \tag{7.1}$$

All arguments are positive and the logarithms are real. The classical digamma difference gives [8, 6]

$$G'(u) = \Phi(-1,1,1+u) = \frac{1}{2}\left\{\psi((u+2)/2) - \psi((u+1)/2)\right\}. \tag{7.2}$$

This relation follows either by separating the even and odd terms in the alternating series or by Hurwitz translation followed by the Laurent expansion at one.

Theorem 7.1 (A complete Euler-weighted Gamma family)**.** *For $r \ge 1$, define the rational number*

$$q_r = -\frac{\partial_s R_{2r}(1,-1)}{(2r)!}. \tag{7.3}$$

Then the ordinary convergent integrals satisfy

$$\begin{aligned} K_{2r} &:= \int_0^1 E_{2r}(u)\Phi(-1,1,1+u)\,\mathrm{d}u \\ &= q_r + \frac{(-1)^{r+1}(2^{2r+1}-1)(2r)!}{(2\pi)^{2r}}\zeta(2r+1), \end{aligned} \tag{7.4}$$

$$\int_0^1 E_{2r-1}(u)G(u)\,\mathrm{d}u = -\frac{K_{2r}}{2r}. \tag{7.5}$$

The first offsets are

$$q_1 = -\frac{1}{2}, \qquad q_2 = \frac{7}{12}, \qquad q_3 = -\frac{53}{30}, \qquad q_4 = \frac{8417}{840}. \tag{7.6}$$

Proof. Apply (6.9) with $d = 2r$, $m = 1$, and $z = -1$. The denominator is $(2r)!$, so

$$K_{2r} = q_r - 2\,\partial_\nu \operatorname{Li}_\nu(-1)|_{\nu=-2r}\,.$$

The identity $\operatorname{Li}_\nu(-1) = (2^{1-\nu} - 1)\zeta(\nu)$, interpreted by continuation at $\nu = 1$, is classical. At a negative even integer $\zeta(-2r) = 0$, hence its derivative there is $(2^{2r+1} - 1)\zeta'(-2r)$. Differentiating the zeta functional equation at that simple trivial zero gives [7]

$$\zeta'(-2r) = \frac{(-1)^r(2r)!}{2(2\pi)^{2r}}\zeta(2r+1).$$

This proves (7.4). The Euler endpoints $E_{2r}(0) = E_{2r}(1) = 0$ and the Appell identity $E'_{2r} = 2rE_{2r-1}$ permit integration by parts in (7.2), proving (7.5) without an omitted boundary constant. The finite polynomial (6.3) gives (7.6) by rational arithmetic. □

The simplest member is the explicit log-Gamma identity

$$\boxed{\int_0^1 \left(u - \frac{1}{2}\right) \log \frac{\Gamma(1 + u/2)}{\Gamma((1+u)/2)}\,\mathrm{d}u = \frac{1}{4} - \frac{7\zeta(3)}{4\pi^2}.} \tag{7.7}$$

We do not attach a global novelty claim to individual low-degree Gamma evaluations; the point is their uniform single-order derivation and compatibility with all higher spectral jets.

7.1 Every positive polygamma order

For integer $m \geq 2$, separation into even and odd terms gives

$$\begin{aligned}\Phi(-1, m, 1+u) &= 2^{-m}\left\{\zeta\left(m, \frac{1+u}{2}\right) - \zeta\left(m, \frac{2+u}{2}\right)\right\} \\ &= \frac{(-1)^m}{2^m(m-1)!}\left\{\psi^{(m-1)}\left(\frac{1+u}{2}\right) - \psi^{(m-1)}\left(\frac{2+u}{2}\right)\right\}.\end{aligned} \tag{7.8}$$

For $1 \leq m \leq d+1$, Theorem 6.1 evaluates the Euler-weighted integral through the one order derivative at $m - d - 1$. For $m \geq d + 2$, there is no vanishing denominator and

$$\int_0^1 E_d(u)\Phi(-1, m, 1+u)\,\mathrm{d}u = \frac{-R_d(m,-1) + 2d!\,\eta(m-d-1)}{\Pi_d(m)}, \tag{7.9}$$

where $\eta(\nu) = (1 - 2^{1-\nu})\zeta(\nu)$ and $\eta(1) = \log 2$. Thus every positive polygamma order has a finite one-center evaluation, including the exceptional orders.

For $E_2(u) = u^2 - u$ and $R_2(s, -1) = s - 1$, the first higher-order members are

$$\int_0^1 (u^2 - u)\Phi(-1, 2, 1+u)\,\mathrm{d}u = 1 + \frac{4}{3}\log 2 + 12\zeta'(-1), \tag{7.10}$$

$$\int_0^1 (u^2 - u)\Phi(-1, 3, 1+u)\,\mathrm{d}u = \log(\pi/2) - \frac{1}{2}, \tag{7.11}$$

$$\int_0^1 (u^2 - u)\Phi(-1, 4, 1+u)\,\mathrm{d}u = \frac{2}{3}\log 2 - \frac{1}{2}. \tag{7.12}$$

For the first two, use (6.9), $\zeta(-1) = -1/12$, $\zeta(0) = -1/2$, and $\zeta'(0) = -\frac{1}{2}\log(2\pi)$. The last is (7.9). Formula (7.8) turns all three into explicit polygamma-difference integrals.

8 Bernoulli selectors and generalized Stieltjes integrals

Fix $N \geq 2$, $a > 0$, and

$$I_N(s;a) = \int_0^1 (B_N(u) - B_N)\zeta(s, a+u)\,\mathrm{d}u. \tag{8.1}$$

For odd $N \geq 3$ the subtracted constant is zero. In all cases the weight has mean $-B_N$. Write

$$b_j = \frac{N!}{(N-j)!}B_{N-j}(1), \qquad 1 \leq j \leq N. \tag{8.2}$$

The resonant endpoint calculation gives

$$\boxed{I_N(s;a) = -\frac{N!}{F_{N-1}(s)}\zeta(s-N,a) + \sum_{j=1}^{N}\frac{b_j a^{j+1-s}}{F_j(s)}.} \tag{8.3}$$

This general-shift, unit-interval form has the same Bernoulli mechanism as the predecessor's [1, 3] theorem [2]. It is included to make the confluence and primitive formulas below directly usable.

We use the Stieltjes convention

$$\zeta(1+\varepsilon, x) = \frac{1}{\varepsilon} + \sum_{m=0}^{\infty}\frac{(-1)^m}{m!}\gamma_m(x)\varepsilon^m, \qquad \gamma_0(x) = -\psi(x). \tag{8.4}$$

For $h \geq 0$, define

$$E_h(\varepsilon) = \prod_{k=1}^{h}(1-\varepsilon/k)^{-1} = \exp\left(\sum_{r\geq 1}\frac{H_h^{(r)}}{r}\varepsilon^r\right). \tag{8.5}$$

Put

$$\begin{aligned} Q_N(\varepsilon;a) &= (-1)^N N E_{N-1}(\varepsilon)\zeta(1-N+\varepsilon, a) \\ &\quad + \sum_{j=1}^{N}\frac{(-1)^j b_j a^j}{j!}E_j(\varepsilon)e^{-\varepsilon\log a}. \end{aligned} \tag{8.6}$$

Proposition 8.1 (All weighted Stieltjes orders at one center)**.** *One has $Q_N(0;a) = -B_N$ and, for every $m \geq 0$,*

$$\boxed{\int_0^1 (B_N(u) - B_N)\gamma_m(a+u)\,\mathrm{d}u = (-1)^m m![\varepsilon^{m+1}]Q_N(\varepsilon;a).} \tag{8.7}$$

This is a finite formula in $\zeta^{(k)}(1-N,a)$ for $0 \leq k \leq m+1$, powers of a and $\log a$, rational Bernoulli numbers, and generalized harmonic numbers.

Proof. At $s = 1+\varepsilon$, $F_j(s)^{-1} = (-1)^j E_j(\varepsilon)/(j!\varepsilon)$. Substituting into (8.3) gives exactly $I_N(1+\varepsilon;a) = Q_N(\varepsilon;a)/\varepsilon$. The compact integral has residue $\int_0^1 (B_N(u) - B_N)\,\mathrm{d}u = -B_N$. Comparison with (8.4), justified uniformly on the compact positive interval, proves both assertions. □

8.1 Three explicit quartic evaluations

For $N = 4$ the weight is $u^2(1-u)^2$ and $b_2 = 2$, $b_3 = 12$, $b_4 = 24$. In particular,

$$I_4(s;a) = -\frac{24\zeta(s-4,a)}{F_3(s)} + \frac{2a^{3-s}}{F_2(s)} + \frac{12a^{4-s}}{F_3(s)} + \frac{24a^{5-s}}{F_4(s)}. \tag{8.8}$$

At $a = 1$ the coefficient generator simplifies to

$$Q_4(\varepsilon;1) = 4E_3(\varepsilon)\zeta(-3+\varepsilon) + E_2(\varepsilon) - 2E_3(\varepsilon) + E_4(\varepsilon).$$

Expanding through degree two, using $\zeta(-3) = 1/120$, yields

$$\int_0^1 u^2(1-u)^2\gamma_0(1+u)\,\mathrm{d}u = 4\zeta'(-3) - \frac{1}{45}, \tag{8.9}$$

$$\int_0^1 u^2(1-u)^2\gamma_1(1+u)\,\mathrm{d}u = -2\zeta''(-3) - \frac{22}{3}\zeta'(-3) + \frac{5}{432}. \tag{8.10}$$

There are no unspecified Stieltjes constants on the right. A different integer resonance of the same complex-order identity gives

$$\boxed{\int_0^1 u^2(1-u)^2\psi'(1+u)\,\mathrm{d}u = \frac{3\zeta(3)}{\pi^2} - \frac{1}{3}.} \tag{8.11}$$

Indeed, at $s = 2+\varepsilon$ the constant term of (8.8) is $-1/3 - 12\zeta'(-2)$, and $\zeta'(-2) = -\zeta(3)/(4\pi^2)$. Every integral here is ordinary and absolutely convergent on a compact interval; no finite-part convention is involved.

9 Joint Stieltjes confluence at the untwisted point

The change of polynomial basis at $z^L = 1$ is algebraic. At $z = 1$ there is a second, analytic issue: the Lerch function, entire in its order before the limit, approaches a Hurwitz function with a pole. The correct limiting object contains an explicit logarithmic term. This section proves a joint holomorphic statement that controls every order derivative at once.

Let $0 < A < B$, and let $w \in L^1([A,B])$ be a fixed complex weight. For polynomial weights, all the finite integrals appearing below can in turn be reduced by endpoint formulas. Set

$$M_w(\mu) = \int_A^B w(x)e^{\mu x}\,\mathrm{d}x, \tag{9.1}$$

$$T_w(\mu,\varepsilon) = \int_A^B w(x)\Phi(e^{-\mu}, 1+\varepsilon, x)\,\mathrm{d}x, \qquad \mu > 0. \tag{9.2}$$

For ε near zero define

$$C(\varepsilon,\mu) = \Gamma(-\varepsilon)\mu^{\varepsilon} + \frac{1}{\varepsilon} \tag{9.3}$$

$$= \frac{1 - \exp\Big((\log\mu+\gamma)\varepsilon + \sum_{r\ge 2}\zeta(r)\varepsilon^r/r\Big)}{\varepsilon}. \tag{9.4}$$

Here γ is Euler's constant, μ^{ε} uses the real logarithm for $\mu > 0$, and the singularity at $\varepsilon = 0$ in each display is removable. The second formula follows from the classical Taylor expansion of $\log\Gamma(1-\varepsilon)$.

Theorem 9.1 (Joint holomorphic confluence and every mixed jet)**.** *The compensated transform*

$$\mathscr{R}_w(\mu,\varepsilon) = T_w(\mu,\varepsilon) - M_w(\mu)C(\varepsilon,\mu) \tag{9.5}$$

extends holomorphically to $|\mu| < 2\pi$, $|\varepsilon| < 1$. *Its exact expansion there is*

$$\begin{aligned}\mathscr{R}_w(\mu,\varepsilon) = &\int_A^B w(x)e^{\mu x}\left(\zeta(1+\varepsilon,x) - \frac{1}{\varepsilon}\right)\mathrm{d}x \\ &+ \sum_{k=1}^{\infty}\frac{(-\mu)^k}{k!}\int_A^B w(x)e^{\mu x}\zeta(1+\varepsilon-k,x)\,\mathrm{d}x.\end{aligned} \tag{9.6}$$

The first bracket is interpreted by its analytic extension at zero. For all nonnegative integers h, m,

$$\begin{aligned}\partial_\mu^h\partial_\varepsilon^m\mathscr{R}_w\Big|_{(0,0)} = &(-1)^m\int_A^B w(x)x^h\gamma_m(x)\,\mathrm{d}x \\ &+ \sum_{k=1}^{h}(-1)^k\binom{h}{k}\int_A^B w(x)x^{h-k}\zeta^{(m)}(1-k,x)\,\mathrm{d}x.\end{aligned} \tag{9.7}$$

The superscript on ζ *differentiates its first argument.*

Proof. The classical Lerch expansion [5, Eq. 25.14.3_1], with $z = e^{-\mu}$, is

$$\Phi(e^{-\mu},1+\varepsilon,x) = e^{\mu x}\left\{\Gamma(-\varepsilon)\mu^{\varepsilon} + \sum_{k=0}^{\infty}\zeta(1+\varepsilon-k,x)\frac{(-\mu)^k}{k!}\right\}. \tag{9.8}$$

Initially $0 < \mu < 2\pi$ and ε avoids zero and the other exceptional integers. Subtract $e^{\mu x}C$ and separate $k = 0$. The result is the integrand version of (9.6).

We record the normal-convergence justification needed for the joint claim. Fix $0 < \delta < 1$ and a compact subdisk $|\mu| \le \rho < 2\pi$. For $k \ge 2$, the Hurwitz Fourier formula at $s = 1+\varepsilon-k$, together with integer translation in the positive argument, gives uniformly for $x \in [A,B]$ and $|\varepsilon| \le \delta$

$$|\zeta(1+\varepsilon-k,x)| \le C\,\Gamma(k+\delta)(2\pi)^{-k} + CD^k \tag{9.9}$$

for fixed C, D. To see the uniformity, reduce the argument to $y \in [0,1]$. The Fourier sums are bounded by $\sum_{n\ge1} n^{-k+\delta}$; their trigonometric factors are bounded, and the gamma factor is controlled on the fixed ε disk. At $y = 0$ use the continuous negative-order limit. Translation back adds finitely many powers $(y+j)^{k-1-\varepsilon}$, bounded by CD^k since $k-1-\delta > 0$. The finite cases and the values $k = 0, 1$ are handled separately after removing the single pole of the $k = 0$ term. The usual gamma-ratio estimate, or its elementary product form on this fixed disk, now makes (9.9) divided by $k!$ a polynomial in k times $(2\pi)^{-k}$, plus an entire-series term $CD^k/k!$.

Consequently the series in (9.6) is normally convergent on compact sub-bidisks and uniformly in $x \in [A,B]$. The factor $e^{\mu x}$ is bounded there. Integration against an L^1 weight preserves normal convergence, so the right side is jointly holomorphic. It agrees with (9.5) in the initial domain and supplies the claimed extension. Cauchy's theorem on smaller disks justifies every spectral derivative as well.

Finally, the coefficient of μ^h in $e^{\mu x}(-\mu)^k/k!$ is $(-1)^k x^{h-k}/(k!(h-k)!)$ when $k \le h$, and zero otherwise. The regular Hurwitz expansion (8.4) gives the $k = 0$ derivative. Multiplying by $h!$ proves (9.7). □

The estimate inside the proof is only used to justify an exact identity and its differentiated limits; no optimization of bounds is part of the result.

9.1 Explicit logarithmic subtraction at every spectral order

Define

$$C_m(\mu) = -m![t^{m+1}] \exp\left((\log\mu + \gamma)t + \sum_{r\geq 2} \zeta(r) t^r/r \right). \tag{9.10}$$

Then $C_m = \partial_\varepsilon^m C(0,\mu)$. With $\ell = \log\mu + \gamma$, the first terms are

$$\begin{aligned} C_0(\mu) &= -\ell, \\ C_1(\mu) &= -\frac{\ell^2 + \zeta(2)}{2}, \\ C_2(\mu) &= -\frac{\ell^3}{3} - \zeta(2)\ell - \frac{2\zeta(3)}{3}. \end{aligned} \tag{9.11}$$

The exact all-order limit is therefore

$$\boxed{\lim_{\mu\downarrow 0} \left\{ \int_A^B w(x)\, \partial_s^m \Phi(e^{-\mu}, s, x)\big|_{s=1}\, \mathrm{d}x - M_w(\mu) C_m(\mu) \right\} = (-1)^m \int_A^B w(x)\gamma_m(x)\, \mathrm{d}x.} \tag{9.12}$$

In particular, zeroth order requires addition of $M_w(\mu)(\log\mu + \gamma)$, not subtraction of that expression. The signs are fixed by the Laurent convention (8.4).

Corollary 9.2 (Mean-zero weights need no logarithmic subtraction)**.** *If $\int_A^B w(x)\,\mathrm{d}x = 0$, then the subtraction term in (9.12) tends to zero for every fixed m, and the uncompensated limit exists with the displayed value. If that mean is nonzero, the uncompensated zeroth-order integral diverges logarithmically.*

Proof. In the mean-zero case $M_w(\mu) = O(\mu)$, whereas $C_m(\mu)$ is a polynomial of degree $m+1$ in $\log\mu$. Their product tends to zero. In the other case $M_w(\mu)$ tends to a nonzero constant and $C_0 = -\log\mu - \gamma$ is unbounded; the remainder has a finite limit by the theorem. □

For $W_{2r}(u) = B_{2r}(u) - B_{2r}$, the mean is $-B_{2r} \neq 0$. Thus the endpoint-flat resonant weights in Theorem 4.2 genuinely require the compensation. Odd Bernoulli weights $B_{2r+1}(u)$ with $r \geq 1$ have mean zero, and their fixed-order Stieltjes limits need none. This is a distinction between endpoint data and mean data, not a contradiction.

9.2 A finite coefficient check at the origin

For $w = 1$ on $[1,2]$, (6.13) gives the original Lerch integral through one polylogarithm derivative. The mass is $M_w(\mu) = (e^{2\mu} - e^{\mu})/\mu$. The extension satisfies

$$\mathscr{R}_w(0,0) = 0, \qquad \partial_\mu \mathscr{R}_w(0,0) = \frac{1}{2}\log(2\pi). \tag{9.13}$$

Indeed the first value is $-\int_1^2 \psi(x)\,\mathrm{d}x = 0$. The second, by (9.7), is $\int_1^2 (-x\psi(x) - \zeta(0,x))\,\mathrm{d}x$. Integration by parts and $\zeta(0,x) = 1/2 - x$ reduce it to $1 + \int_1^2 \log\Gamma(x)\,\mathrm{d}x = \frac{1}{2}\log(2\pi)$, using the classical Raabe integral. This verifies both the twist coordinate and the sign of the compensator.

10 Anchored primitives in the Hurwitz argument

Endpoint closure also survives repeated integration in the shift. For this section P and L are fixed; abbreviate $T(s;a) = \mathcal{T}_P(z,s;a,L)$. Let $a, a_0 > 0$, and let $q \geq 1$. The Pochhammer symbol in the next display is rising: $(s-q)_q = (s-q)(s-q+1)\cdots(s-1)$.

Theorem 10.1 (All anchored shift primitives)**.** *Define meromorphically in s*

$$J_q(s;a,a_0) = \frac{(-1)^q}{(s-q)_q}\left\{T(s-q;a) - \sum_{j=0}^{q-1}\frac{(a-a_0)^j}{j!}\partial_a^j T(s-q;a_0)\right\}. \tag{10.1}$$

Then

$$\partial_a^q J_q(s;a,a_0) = T(s;a), \qquad \partial_a^j J_q(s;a,a_0)\Big|_{a=a_0} = 0 \quad (0 \leq j < q). \tag{10.2}$$

For $z \neq 1$ all apparent spectral singularities of J_q are removable. For $z = 1$ the only possible pole is at $s = 1$, with

$$\operatorname{Res}_{s=1} J_q(s;a,a_0) = \frac{(a-a_0)^q}{q!}\int_0^L P(u)\,\mathrm{d}u. \tag{10.3}$$

Consequently, if $M = \int_0^L P(u)\,\mathrm{d}u$, the coefficient

$$\mathcal{A}_{m,q}(a,a_0) = (-1)^m m![\varepsilon^m]\left\{J_q(1+\varepsilon;a,a_0) - \frac{M(a-a_0)^q}{q!\varepsilon}\right\} \tag{10.4}$$

is the unique qth anchored primitive of $\int_0^L P(u)\gamma_m(a+u)\,\mathrm{d}u$.

Proof. Repeated argument differentiation gives $\partial_a^q T(s-q;a) = (-1)^q(s-q)_q T(s;a)$. The Taylor subtraction proves the anchors, and proves (10.2) away from the exceptional orders. Alternatively the resulting function is the ordinary iterated integral

$$J_q(s;a,a_0) = \frac{1}{(q-1)!}\int_{a_0}^a (a-v)^{q-1}T(s;v)\,\mathrm{d}v, \tag{10.5}$$

with the standard oriented integral if $a < a_0$. It gives analytic continuation in s and uniqueness with the anchors immediately.

The cancellation is also visible in the finite formula. For $z \neq 1$, $\Phi(z,-k,a)$ is a polynomial in a of degree at most k: expand $(n+a)^k$ in the convergent series when $|z| < 1$, and then continue to the unit circle away from one. At each zero $s = 1,\ldots,q$ of the Pochhammer factor, the Taylor remainder therefore vanishes. For $z = 1$, $\zeta(-k,a) = -B_{k+1}(a)/(k+1)$. The zeros $s = 2,\ldots,q$ still cancel; at $s = 1$ the degree-q term gives precisely (10.3). The potential pole at $s = q+1$ of the shifted $T(s-q)$ has residue independent of a, so the zeroth Taylor term cancels it. This accounts for every apparent singularity of (10.1).

Finally apply (8.4) to the ordinary iterated integral (10.5). Coefficient extraction gives (10.4), including all anchors. □

For $q = 1$ the Stieltjes formula is particularly transparent. With $z = 1$ and $T(s;a) = \int_0^L P(u)\zeta(s,a+u)\,\mathrm{d}u$,

$$\mathcal{A}_{m,1}(a,a_0) = \frac{(-1)^{m+1}}{m+1}\left\{\partial_s^{m+1}T(0;a) - \partial_s^{m+1}T(0;a_0)\right\}. \tag{10.6}$$

At $m = 0$, Lerch's formula for $\zeta'(0, x)$ reduces it to

$$\mathcal{A}_{0,1}(a, a_0) = -\int_0^L P(u) \log \frac{\Gamma(a+u)}{\Gamma(a_0+u)}\, \mathrm{d}u. \tag{10.7}$$

At $m = 1$ it is one half of the corresponding difference of weighted $\zeta''(0, x)$ integrals. For selector weights, the endpoint identities express these further in finitely many negative-order zeta derivatives. This is an exact anchored calculus; an unspecified additive polynomial would lose the normalization required by the source programme.

11 A related open coefficient problem and normalization corrections

The literature search located a concrete complementary question in Pokaz, Ribičić Penava, and Simsek, arXiv:2609.13796v1 [11]. On page 9, the authors ask for the coefficients in a finite Hurwitz–Lerch reduction of a nested series. Their examples cover the first few values of a discrete parameter. The complete answer is a binomial row. This is an elementary partial-fraction resolution of the question as posed in that version, not a claim that the partial-fraction method itself is new.

Put $n = v - 1 \geq 1$, $a = x - 1 > 0$, and define

$$h_m^{[n]} = \sum_{m_1+\cdots+m_n=m} \prod_{j=1}^{n} j^{-m_j}, \qquad Z_v(s; a) = -\frac{(s)_v}{(v-1)!} \sum_{m\geq 0} \frac{h_m^{[v-1]}}{(a+m)^{s+v}}. \tag{11.1}$$

Here $(s)_v$ is rising and $\Re(s + v) > 1$ initially. The change $m_1 = k_1$, $m_j = k_j - k_{j-1}$, and $m_{v-1} = m - k_{v-2}$ identifies this with Definition 1 of [11], with $Z_v(s; a) = Z(s; a + 1, v)$ in that paper's notation.

Theorem 11.1 (The coefficient question in [11])**.** *For every $v \geq 2$ and $1 \leq j \leq v - 1$, set*

$$\boxed{\alpha_j = (-1)^{j-1} \binom{v-1}{j}.} \tag{11.2}$$

Then, identically in s by continuation,

$$\boxed{Z_v(s; a) = -\frac{(s)_v}{(v-1)!} \sum_{j=1}^{v-1} (-1)^{j-1} \binom{v-1}{j} \Phi(1/j, s+v, a).} \tag{11.3}$$

The $j = 1$ term is $(v - 1)\zeta(s + v, a)$, as required in the source question. The function $Z_v(s; a)$ is entire in s.

Proof. The generating function for the finite-composition coefficients is

$$\sum_{m\geq 0} h_m^{[n]} t^m = \prod_{j=1}^{n} (1 - t/j)^{-1}.$$

The coefficient of $(1 - t/j)^{-1}$ in its partial-fraction expansion is

$$\prod_{\substack{1\leq k\leq n \\ k\neq j}} (1 - j/k)^{-1} = \frac{n!/j}{(-1)^{j-1}(j-1)!(n-j)!} = (-1)^{j-1} \binom{n}{j}.$$

Both sides have the same simple poles and principal parts and tend to zero at infinity, so the partial-fraction identity is exact. Comparing coefficients gives

$$h_m^{[n]} = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} j^{-m}. \tag{11.4}$$

These coefficients are bounded as $m \to \infty$; hence the defining Dirichlet series is absolutely convergent for $\Re(s+v) > 1$. Substitute (11.4) and interchange the finite sum to obtain (11.3). Every term with $j \geq 2$ is entire. The sole Hurwitz pole at $s = 1 - v$ is canceled by the simple zero of $(s)_v$. Thus Z_v is entire. □

11.1 Every jet at the canceled Hurwitz pole

The same harmonic factor as in the selector calculus appears at $s_0 = 1 - v$. Write $n = v - 1$ and $D_n(\varepsilon) = \prod_{k=1}^{n}(1 - \varepsilon/k) = E_n(\varepsilon)^{-1}$. Then (11.3) is equivalent near s_0 to

$$Z_v(1 - v + \varepsilon; a) = (-1)^v D_{v-1}(\varepsilon) \Bigg[(v-1) + \varepsilon \left\{ (v-1) \sum_{m \geq 0} \frac{(-1)^m \gamma_m(a)}{m!} \varepsilon^m + \sum_{j=2}^{v-1} \alpha_j \Phi(1/j, 1+\varepsilon, a) \right\} \Bigg]. \tag{11.5}$$

This convergent local identity proves every spectral derivative by finite coefficient extraction. In particular,

$$Z_v(1 - v; a) = (-1)^v (v-1), \tag{11.6}$$

$$Z_v'(1 - v; a) = (-1)^v \left\{ (v-1)(\gamma_0(a) - H_{v-1}) + \sum_{j=2}^{v-1} \alpha_j \Phi(1/j, 1, a) \right\}. \tag{11.7}$$

For example $Z_3'(-2; 1) = 3 - 2\gamma + 2\log 2$. Thus the coefficient resolution also supplies exact Stieltjes and logarithmic evaluations, not just an interpolation of the first few coefficient rows.

11.2 Normalization fixed by the defining series

The definition (11.1), not an inconsistent later normalization, is used throughout. Direct Mellin integration of that definition gives, for $a > 0$ and $\Re(s+v) > 1$,

$$\boxed{Z_v(s; a) = -\frac{1}{(v-1)!\,\Gamma(s)} \int_0^\infty \frac{t^{s+v-1} e^{-at}}{\prod_{j=1}^{v-1}(1 - e^{-t}/j)}\, \mathrm{d}t.} \tag{11.8}$$

At poles of $\Gamma(s)$ this is interpreted by its entire reciprocal. To prove it, expand each geometric denominator, integrate $t^{s+v-1} e^{-(a+m)t}$, and use $(s)_v = \Gamma(s+v)/\Gamma(s)$ by continuation.

In the inspected version of [11], equation (17) on page 11 has the prefactor $-(v-1)!/\Gamma(s)$ instead. Relative to its own Definition 1 this differs by $((v-1)!)^2$. The subsequent $v = 2$ integral on that page also has an incorrect sign and gamma factor. Its corrected form is

$$\int_0^\infty \frac{t^{s+1} e^{-at}}{1 - e^{-t}}\, \mathrm{d}t = \Gamma(s+2)\zeta(s+2, a), \qquad a > 0, \quad \Re s > -1. \tag{11.9}$$

This follows termwise from a positive kernel when s is real; a negative right side in that domain cannot be correct. Both source pages were checked in the rendered PDF, not only in extracted text.

One can also settle the sign of the polynomial interpolation without a contour orientation convention. Retain the source generating function

$$F(t,x;v)=\frac{t^v e^{xt}}{\prod_{j=0}^{v-1}(e^t-j)}=\sum_{k\ge 0}\ell_k(x;v)\frac{t^k}{k!}. \tag{11.10}$$

Algebra gives

$$F(-t,a+1;v)=-\frac{t^v e^{-at}}{(v-1)!\prod_{j=1}^{v-1}(1-e^{-t}/j)}.$$

Therefore (11.8) is $\Gamma(s)^{-1}\int_0^\infty t^{s-1}F(-t,a+1;v)\,\mathrm{d}t$. Subtracting a finite Taylor polynomial at zero continues this Mellin integral. Its residue at $s=-k$ is $(-1)^k\ell_k(a+1;v)/k!$, whereas $\Gamma(s)^{-1}\sim(-1)^k k!(s+k)$. Hence

$$\boxed{Z_v(-k;a)=\ell_k(a+1;v),\qquad k\ge 0.} \tag{11.11}$$

The minus sign in equation (19) of [11] is incompatible with Definition 1. Already $v=2$, $k=1$ gives $Z_2(-1;a)=1$ and $\ell_1(a+1;2)=1$. This correction is scoped to that source and normalization; it is not an allegation about the ProveIt manuscript.

12 Source audit, verification, and integration status

12.1 What was inspected

The canonical manuscript README and the incoming-directory metadata were read through the connected GitHub tools on 11 October 2026. Selected harmonic-tail sections were read at commit `47f2d0da4f1c0e19f9ec8aecb39112756f167d82`. A later repository search used a newer indexed commit. These observations are not represented as a single frozen snapshot of a changing `main` branch.

The precise mathematical predecessor for this article is the 1,702-line standalone TeX source of *Affine Gamma Polarization*, obtained from the user's Library. Its sections on Lerch endpoints, Bernoulli selectors, Stieltjes coefficients, and further questions were inspected. Its SHA-256 and Library version are recorded in `source_manifest.json`. The GitHub incoming directory independently lists the corresponding named archive, but its binary contents were not extracted in this run. We do not assert a byte-for-byte comparison between the Library TeX and every member of that archive. No external source PDF or predecessor TeX is redistributed in this package.

For the external coefficient question, the defining series, the open question, the Mellin normalization, and the interpolation statement in arXiv:2609.13796v1 were checked in both the HTML and the rendered PDF. The relevant printed pages are 3, 9, 11, and 12. The corrections in Section 11 are tied to that version's Definition 1. They do not silently change its definition to match a later, inconsistent prefactor.

12.2 Corrections and limitations

There is no false theorem identified in the reviewed selector portion of the incoming ProveIt report. Its open question genuinely leaves room for the classification proved here. Three cautions are nevertheless necessary when integrating the continuation.

First, one cannot deform a double-flat Bernoulli selector into a nonresonant double-flat single-order polynomial. The obstruction is Theorem 4.2, not a failure to find the appropriate normalization. The exact replacement is the finite mixture (5.1). Second, a fixed-degree Frobenius–Euler selector cannot have a nonconstant finite scalar limit at resonance; the coefficientwise result in Section 5 explains why singular coefficients in the mixture must cancel collectively. Third, the limit $z \to 1$ cannot be substituted into a spectral jet at $s = 1$ without its pole/contact subtraction. Equations (9.3)–(9.12) supply the complete subtraction, including Euler's constant and all higher zeta terms.

The external corrections are independent: relative to the defining series, equation (17) of [11] has an erroneous factorial factor; its displayed $v = 2$ integral has the wrong sign and gamma factor; and equation (19) has the wrong interpolation sign. The replacement formulas (11.8), (11.9), and (11.11) are derived from the definition and checked independently. This is a scoped audit, not a claim to have verified every formula in that paper or in the ProveIt volumes.

12.3 Proof status and dependence

Result	Scope and principal justification
Selector classification	Every integer length, $a > 0$, and $0 < \lvert z\rvert \le 1$; endpoint identity and functional independence.
Endpoint obstruction	Every degree; Eulerian interlacing and the classical Appell boundary relation.
Resonant support collapse	Exact identity at every even degree; Frobenius–Euler endpoint reconstruction.
Spectral and Stieltjes jets	Every jet order; analytic removability, finite coefficient extraction, and normal convergence.
Anchored primitives	Every primitive order; Taylor subtraction and ordinary iterated integration.
External coefficient question	Every $v \ge 2$; explicit rational partial fractions, not extrapolation of finite rows.
Arithmetic independence	Not asserted after specializing s. Functional independence over $\mathbb{C}(s)$ is a different theorem.

The Frobenius–Euler basis theorem is already present in Kim and Kim [3]; the endpoint transform uses classical Hurwitz/Lerch calculus. Integrals of the Hurwitz zeta function, log-Gamma functions, and negative-order polygamma functions have a substantial prior literature, including Espinosa and Moll [9, 10]. The examples here are consequences in a specified selector system, not claims that weighted Gamma integrals or their general method were previously unknown. Likewise, the coefficient solution in Section 11 is a complete answer to a stated question, but uses elementary and classical partial fractions. Global bibliographical novelty remains unverified.

12.4 Exact checks

The scripts use SymPy 1.14.0 and rational arithmetic. The original selector suite contains 667 assertions. The separately implemented coefficient-problem suite contributes 302 more, for a total of **969 finite exact assertions**. Six of these are mutation controls which deliberately insert an incorrect expression and verify its rejection. The checks include the following independent constructions.

Selector-suite family	Assertions
Frobenius–Euler endpoint derivatives	180
Appell derivatives and endpoint reconstruction	74
Bernoulli resonant derivatives	90
Eulerian endpoint identities, factorial values, consecutive gcds	29
Algebraic twist and exact support strata	16
Finite-mixture identities and endpoint multiplicities	25
Removable-integer numerator cancellation	135
Generalized-harmonic coefficient recurrence	112
Printed Stieltjes/trigamma identities	4
Mutation controls	2
Selector subtotal	667
External rational partial fractions	10
Nested-composition coefficients versus binomial rows	210
Defining generating series versus continued integer values	78
External normalization/sign mutation controls	4
Combined total	969

The interpolation tests compute the coefficients of the source function $F(t, 2; v)$ by rational power-series convolution. The other side is computed from continued zeta/Lerch values, with the canceled pole handled separately. Thus the sign tests do not simply evaluate two copies of the same Mellin formula. The external coefficient tests also generate the nested coefficients by multiplying geometric series rather than by using their binomial answer.

12.5 Numerical diagnostics

The two diagnostic suites use mpmath 1.3.0 at 75 decimal digits of working precision. There are **33 comparisons**: 27 for the selector calculus and six for the external coefficient and Mellin identities. The largest recorded scaled discrepancy is approximately 3.075×10^{-36}. This comes from the independent quartic first-Stieltjes integral, computed with a finite Gauss–Legendre rule; its acceptance tolerance is 10^{-32}. Individual absolute and scaled discrepancies and all tolerances are recorded in the result JSON files. Working precision is not an accuracy certificate.

The comparisons include direct geometric sums against endpoint formulas, removable-order limits, an order derivative, ordinary Euler-weighted digamma and log-Gamma quadratures, three quadratic polygamma identities, the quartic Stieltjes and trigamma formulas, compensated Lerch expansions, a mixed confluence derivative, and an anchored primitive. The external Mellin comparisons use an independently accumulated nested-coefficient series; its constant tail is summed by Hurwitz zeta and its remaining geometric tail is truncated. The canceled-pole jet is checked by Cauchy differentiation on a small circle, avoiding direct multiplication of a rounded zero by a pole.

These are floating-point diagnostics, not interval-certified enclosures. They can detect index, sign, and implementation errors. They do not replace any all-order proof, prove numerical period independence, or certify global novelty. No theorem prover has been used.

12.6 Integration proposal

Retain this article and its scripts as a separate continuation report, with the ledger marking the integer-length version of the source's questions 4 and 5 as resolved in the stated functional sense. The

rational-endpoint, noninteger-length question remains separate. The primitive and joint-jet theorems advance question 6 but do not supply its requested compact degree/order bivariate generating function or a complete Barnes-Gamma classification.

Every label in this source uses the `pls:` prefix. The package includes a short integration fragment, an explicit source manifest, reproduction instructions, and exact/numerical receipts. No repository files have been edited. In particular, this delivery does not change the conjectural status of S_6, the revised S_8, or the general unequal-tail optimal-prefix problem.

13 Further research questions

1. Prescribed spectral support with higher endpoint vanishing. Classify polynomial weights selecting two or more prescribed shifted orders while vanishing to a specified order at both endpoints. Theorem 4.2 settles the one-coordinate obstruction. For larger support, the exact problem is the rank of an endpoint matrix of Frobenius–Euler polynomials. Determine its exceptional algebraic twists and prove minimal support in a declared degree range. Counting equations alone does not prove the necessary minors are nonzero.

2. Rational interval lengths and cyclotomic coordinates. For noninteger rational L, translation no longer returns the same Hurwitz argument modulo a finite prefix. Determine a complete selector basis with a finite vector of rational residue-class arguments, then use roots-of-unity filters to convert it to polylogarithms or Dirichlet L-functions. The coefficient field and the permitted finite remainder must be fixed before stating uniqueness. Integer L is completely covered here; this different endpoint geometry is not.

3. Coalescing geometric poles. Replace the distinct nodes $1, 1/2, \dots, 1/(v-1)$ in Section 11 by arbitrary nodes with multiplicities. Partial fractions predict finite derivatives in the Lerch twist, while the collision limit is governed by divided differences. Prove a stable all-jet confluence theorem and determine which normalizations give particularly simple Gamma-product or Stieltjes interpolation values. Keep the original defining series as the normalization anchor.

4. Resonant roots of unity and simultaneous degenerations. At $z^L = 1$ with $z \neq 1$, spectral support collapses but there is no Hurwitz pole. Combine the exact support law with finite Fourier transforms to distinguish this pure endpoint resonance from the additional logarithmic contact at $z = 1$. Study simultaneous variation of the interval shift, twist, and spectral order, with an explicit finite subtraction at every singular face.

5. A compact degree–Stieltjes generating identity. The coefficient rules in Sections 6 and 8 compute every fixed degree and every Stieltjes order. Seek a bivariate generating identity that retains a single spectral center and exposes the complete-harmonic coefficients without a separate finite sum at each degree. Determine which anchored primitives close in a specified finite family of multiple-Gamma or zeta-derivative functions. This is the remaining substantive part of the predecessor's question 6.

6. Nonlinear Gamma sectors with exact selectors. The incoming affine Gamma square-sector formula motivates the beta and Bernoulli weights used here. Transport multi-coordinate

selectors back through that normal-form construction, retaining every endpoint term. Seek complete cubic or quartic derivative-sector identities, not only diagonal squares or low-degree examples. A useful outcome would be a generating formula whose Hurwitz/Lerch transform has a provably minimal, prescribed functional spectrum.

7. The remaining harmonic-tail and polylogarithm questions. The general unequal-even-tail optimal-prefix conjecture concerns nonvanishing of specific rational coefficient determinants and is not implied by the independence lemma here. Find an exact representation of those determinants or a recurrence proving nonvanishing. Separately, the S_6 and revised S_8 period candidates require exact parameter identities with controlled boundary constants. A failed relation search, or an extremely small residual, does not settle either problem.

8. Proof-carrying symbolic integration. Formalize the endpoint map, its basis inversion, the rational partial-fraction theorem, and the finite harmonic coefficient recurrence in a proof assistant. These finite algebraic cores can be checked independently of the harder analytic continuation library. Then add normal convergence, argument differentiation, and anchored primitive uniqueness. The resulting compiler should emit both an identity and a machine-checkable proof of its domain and normalization.

Conclusion

Paired endpoint derivatives turn an integral-classification problem into finite interpolation, while functional independence proves that no hidden order has been discarded. Resonance changes the rank and therefore the degree of the selector; it also explains why a finite mixture, rather than scalar normalization, is needed to recover a nonconstant Bernoulli weight. Complete-harmonic coefficients and a single explicit logarithmic counterterm then connect this structure to Gamma integrals, all Stieltjes jets, and anchored primitives. The same attention to finite coefficients resolves the external nested-series question and fixes its normalization. All claims retain their precise algebraic, analytic, and computational scopes.

References

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[2] *Affine Gamma Polarization*, ProveIt research continuation, 11 October 2026. Standalone TeX, Library version 1, 1,702 lines. Sections 7, 9, and 11 are the principal predecessor material used here. Named incoming archive: `ProveIt_Affine_Gamma_Polarization_2026-10-11.zip`. https://github.com/VladimirReshetnikov/ProveIt/tree/main/docs/incoming. The standalone-source hash is recorded in `source_manifest.json`.

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[11] Dora Pokaz, Mihaela Ribičić Penava, and Yilmaz Simsek, *Certain families of series involving Hurwitz and Lerch zeta functions*, arXiv:2609.13796v1, 12 September 2026. The open coefficient question is on page 9; the corrected normalization and interpolation statements are on pages 11–12. Version-specific corrections are proved in Section 11. https://arxiv.org/abs/2609.13796v1.